

Social Systems Engineering

Cross-disciplinary integration across INCOSE and organizational SE practice

The SSE WG's highest-value contribution is serving as the socio-technical context layer that activates and calibrates requirements and governance across all engineering disciplines. Every system that interacts with humans is a socio-technical system. SSE's integration mandate follows directly from this fact.

This proposal identifies four structural integration points: stakeholder context capture, socio-technical norms as requirements inputs, design and validation governance, and continuous run-time review. Each is mapped to INCOSE WG activity and to the SE project lifecycle.

Foundation: SSE in nested ecosystems

Every artificial system produced through systems engineering exists within a nested hierarchy of ecosystems operating on fundamentally different timescales. The planetary ecosystem sets absolute constraints that shift over centuries and millennia — Rockström's planetary boundaries are the current best formalization of these limits. The social ecosystem — institutions, governance structures, cultural values, and community resilience — adapts over years and decades. Artificial systems, the outputs of SE practice, cycle in months to years. The root cause of the tactical/strategic tension is not a values problem. It is an architectural one: artificial systems are structurally optimized for their own timescale and have no built-in mechanism to sense or respond to slow-variable signals from the outer layers. A market that does not price ecological carrying capacity is not immoral — it is architecturally decoupled from the planetary layer.

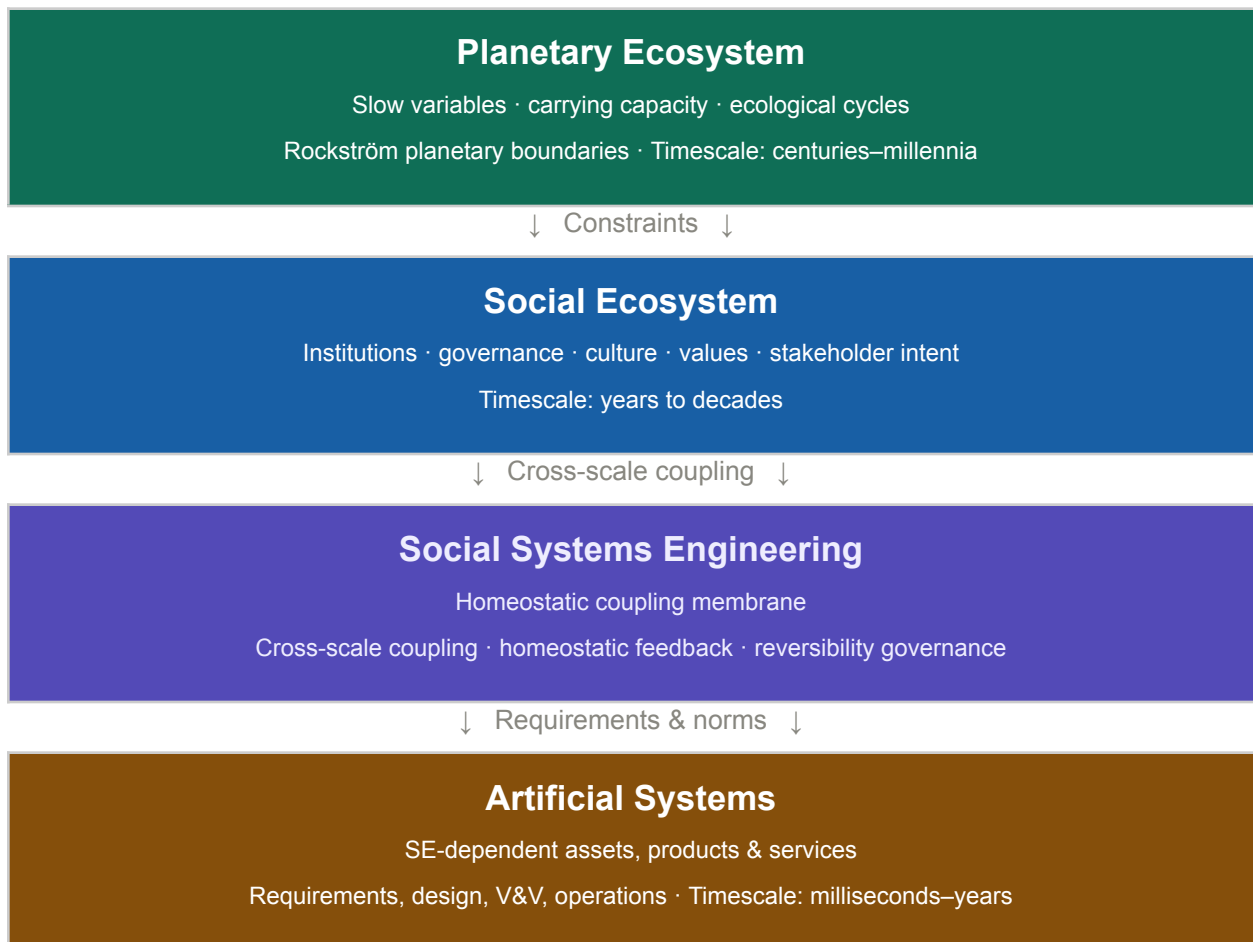
Social Systems Engineering is the discipline that must live in the coupling zone between the social and artificial layers. Not inside the social ecosystem, and not inside the technical layer — at the membrane between them. Its role is to engineer the interfaces, feedback mechanisms, and governance structures that keep fast-cycle artificial systems in coherent relationship with the slow-variable constraints above them. Three functions define this membrane role: cross-scale coupling ensures planetary and social signals actually reach design decisions; homeostatic feedback means designing artificial systems to maintain regulated states rather than to optimize toward a single objective at a single timescale monotonically; and reversibility governance makes irreversibility a first-class design constraint, so tactical decisions cannot silently foreclose strategic options.

The science underlying this framework is established. Ashby's Law of Requisite Variety establishes that a governance system must be at least as complex as the disturbances it faces, which is why simplified SE processes cannot govern complex socio-technical-ecological systems. Beer's Viable System Model provides the structural requirements for systems that remain viable across changing environments. Holling's panarchy theory explains why resilience requires adaptive capacity across scales, not just optimization within one. The modern industrial era's polycrisis results precisely from optimizing fast-cycle artificial systems in ways that

attenuate feedback from slow-cycle planetary variables. Reconnecting those feedback loops is not supplemental to systems engineering. It is the condition of SE's long-term relevance as a discipline.

The tactical/strategic tension is structural, not motivational. Markets do not price ecological carrying capacity. Project timelines do not contain civilizational feedback loops. SSE's membrane role is to engineer the coupling mechanisms — semantic gateways, sustainability-grounded requirements, reversibility constraints — that make slow signals legible inside fast systems.

Figure 0 · SSE as homeostatic coupling membrane in nested ecosystems



Natural ecosystems operate through nested adaptive cycles across temporal scales. This is the essential insight of Holling's panarchy. The planetary layer — biogeochemical cycles, climate dynamics, evolutionary selection pressures — operates on century-to-millennial timescales. It sets the envelope within which social and institutional systems can exist. The social layer — governance structures, cultural norms, legal commons, meaning systems — operates on decadal-to-century timescales. It translates planetary signals into human-intelligible norms and constraints. Artificial systems — technology, markets, infrastructure, industrial processes — operate on the daily-to-annual timescale of human action. They are where we do things, but they are not where the governing logic lives.

The fundamental asymmetry of this architecture is critical: **constraint flows downward, and signals must flow upward**. Planetary limits constrain social possibility; social norms govern design intent; engineering intent shapes what artificial systems do. Signals from operational reality must propagate back upward — operational data informs social learning, ecological signals inform institutional response. When this signal propagation is blocked or distorted, the system loses its homeostasis. That is precisely what the modern industrial era has done: it optimized fast-cycle artificial systems in ways that attenuated the feedback signals from slow-cycle planetary variables. The result is the polycrisis we are now navigating.

Social Systems Engineering is the discipline that must live in the coupling zone, not inside the social ecosystem (that is, governance, law, anthropology), not inside the technical layer (that is, engineering in the conventional sense), but at the membrane between them. Its job is to engineer the interfaces, feedback mechanisms, reversibility architecture, and governance structures that ensure fast-cycle artificial systems remain in coherent relationship with slow-cycle social and planetary constraints. Three functions define this membrane role:

Cross-scale coupling means ensuring that planetary and social signals actually reach the design decisions made within artificial systems. This is harder than it sounds. Markets, for instance, are artificial systems that are structurally capable of ignoring externalities — they couple to short-term price signals but are architecturally decoupled from slow-variable planetary costs. SSE must engineer the coupling mechanisms that bring the planetary signals in: semantic gateways, requirements traceability to sustainability constraints, and ontological grounding that makes "carrying capacity" as legible to an industrial system as "throughput."

Homeostatic feedback means designing artificial systems to maintain regulated states rather than to monotonically optimize. Natural ecosystems maintain homeostasis through redundancy, modularity, and negative feedback loops operating across multiple timescales simultaneously. Artificial systems, by default, optimize for a single objective at a single timescale — which is a recipe for precisely the tactical-at-the-expense-of-strategic failure mode you are asking about. The science here draws on cybernetics (Beer's Viable System Model is directly applicable), complexity theory, and ecological resilience theory (Holling, Walker). Ashby's Law of Requisite Variety is foundational: the regulatory capacity of a governance system must be at least as complex as the disturbances it faces. Artificially simplified systems cannot govern complex natural ones.

Reversibility governance is perhaps the most actionable principle for the tactical/strategic tension. The reason tactical decisions sacrifice strategic needs is almost always because they lock in irreversible states — physical infrastructure, regulatory capture, organizational path dependence, and ecological degradation. If reversibility is a first-class design constraint on artificial systems, the temporal tension dissolves significantly. A tactical solution that remains reversible cannot permanently compromise a strategic need. This maps directly onto the reversibility taxonomy in the RDSG*, and it is not coincidental that nature's own resilience depends on the α (release) and Ω (reorganization) phases of the adaptive cycle, which are precisely the moments of reversibility and reconfiguration that most artificial systems structurally suppress.

The tactical/strategic reconciliation, properly framed, is not a trade-off. It is a design problem with science. The failure mode is optimizing the fast cycle in ways that erode the slow variables that make the fast cycle viable. The correction is to treat slow-variable preservation as a hard constraint on fast-variable optimization — not a competing objective but a feasibility boundary. Planetary boundaries (Rockström's framework) provide the outer constraint envelope. Ostrom's design principles for commons governance provide the social-layer institutional architecture. Holonic systems design (domain-specific models) provides the architectural pattern

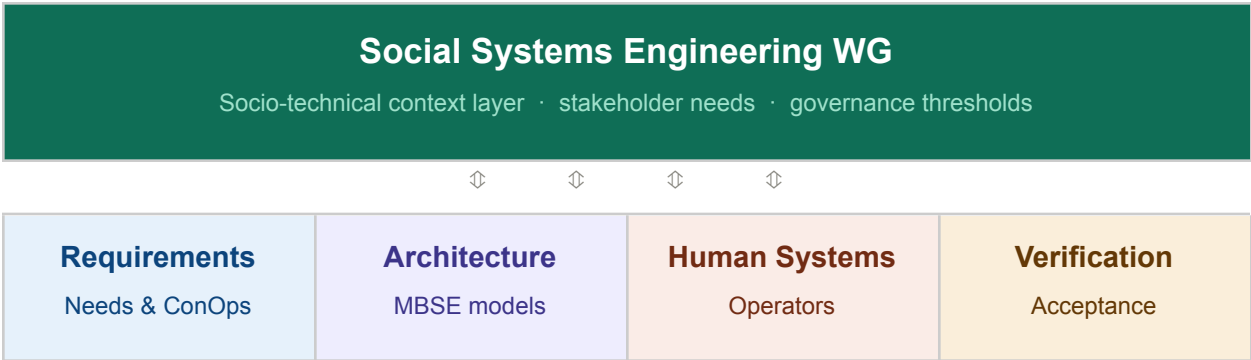
that allows local autonomy (tactical responsiveness) within nested systemic constraint (strategic coherence). Each holon satisfies its own operational imperatives while remaining governed by the holarchy above it. That is exactly how healthy ecosystems function.

The implications for applying science to artificial systems specifically are these: design for carrying capacity respect rather than efficiency alone; design for adaptive capacity rather than optimization convergence; design for graceful degradation under stress rather than catastrophic brittleness; and design cross-scale feedback propagation as a structural requirement, not an afterthought. The Planet > People > Profit hierarchy is not merely a values statement; it is the correct topological ordering of the constraint hierarchy itself. Planetary integrity constrains social possibility; social coherence constrains economic viability. Artificial systems that invert this ordering are not just unethical; they are literally thermodynamically incoherent over any sufficiently long timescale.

SSE in the INCOSE ecosystem

The ecosystem foundation above explains why SSE’s integration role is non-negotiable in principle. The diagram below shows how it maps in practice: SSE is positioned as a cross-cutting context layer with bidirectional relationships to four core INCOSE working group clusters. Technical WGs provide the architectural vocabulary that gives SSE precision; SSE provides the socio-technical framing that makes technical outputs trustworthy and durable.

Figure 1 · INCOSE WG integration model



Integration across the SE project lifecycle

Three SSE activity zones span the full SE lifecycle. The design-time zone covers needs capture, requirements input, and socio-technical validation. The run-time zone covers continuous review of whether deployed systems remain aligned with stakeholder intent. This division reflects a fundamentally different governance relationship with the system before and after deployment.

Figure 2 · SE lifecycle integration touchpoints

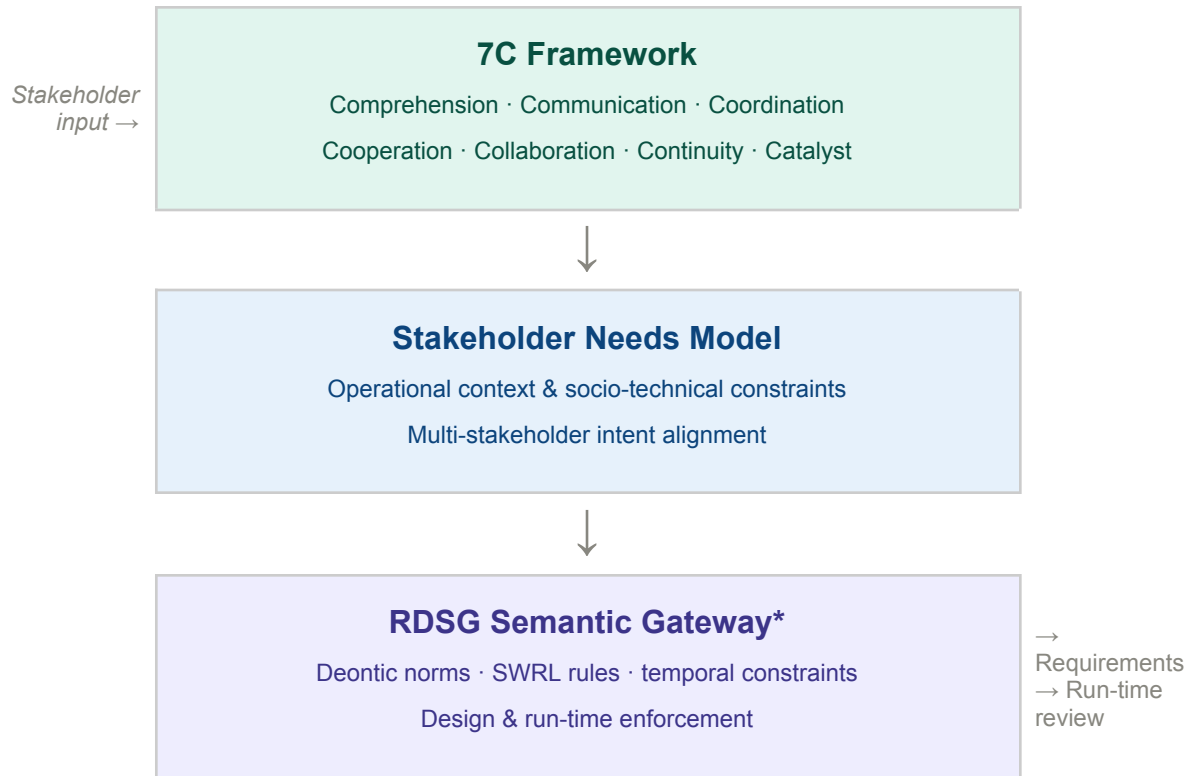


The 7C framework as a requirements pipeline

The seven C-activities are not communication principles alone — they are a structured pipeline that produces artifacts feeding requirements governance. Comprehension and Communication generate operational context; Coordination and Cooperation surface competing stakeholder objectives; Collaboration, Continuity, and Catalyst define what must persist and adapt. Together,

they produce a Stakeholder Needs Model that feeds the RDSG Semantic Gateway, where constraints are encoded as deontic norms enforced across design and run-time.

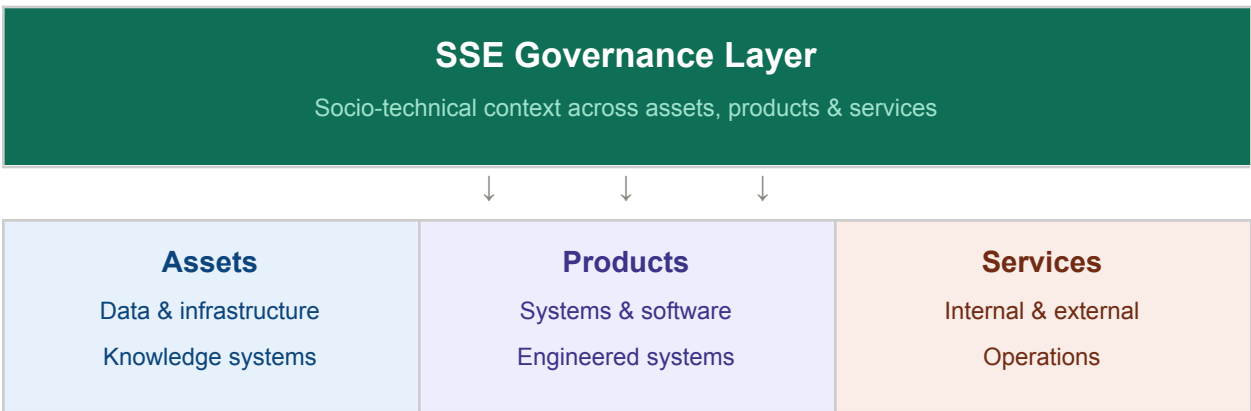
Figure 3 · 7C framework to semantic governance pipeline



Organizational scope of SSE governance

In organizations that rely on systems engineering, SSE governance applies across the full portfolio of assets, products, and services — internal or external, involving one engineering discipline or many. The governance layer does not command these systems; it provides the continuous socio-technical context that keeps their designs aligned with the communities they serve across the full lifecycle.

Figure 4 · Organizational scope of SSE governance



Recommended integration actions

Five actions represent the minimum viable path to establishing SSE as a structural participant in INCOSE's engineering practice:

1. Cross-WG liaison roles

Embed SSE representatives in Requirements, Human Systems Integration, and Architecture WGs. This is the single highest-leverage structural change and requires no new WG formation or dedicated funding.

2. Map 7C to ISO 15288 lifecycle stages

Demonstrate concretely where each C-element activates and what artifacts it produces or consumes. This creates a shared integration grammar between SSE and technical WGs.

3. Define socio-technical threshold indicators

In collaboration with each technical WG, define what socio-technical drift looks like in their domain and at what level it triggers a governance review. These become SSE's run-time contribution.

4. Pilot a joint STPA/UCA workshop

Run a single session with one technical WG demonstrating how SSE methods surface unsafe control action (UCA) candidates that purely technical STPA analysis misses. This is the strongest demonstration of additive value.

5. RDSG v2.0 as reference implementation*

The Requirements-Driven Semantic Gateway Architecture has SSE integration points structurally defined. Use it as a concrete demonstration vehicle for Figure 3 and as a pilot for broader cross-WG adoption. *This is an initiative that I've been working on for 2026.*

6. Establish connections to natural systems science

Formally ground SSE's governance claims in Ashby's Law of Requisite Variety, Beer's Viable System Model, Holling's panarchy theory, and Rockström's planetary boundaries framework. These transform SSE's claims from normative preferences into scientific necessities. Identifying three or four academic co-leads who can formalize this grounding in a joint WG paper is the highest-leverage academic investment the group can make in year eight.